

Finite Quantum Dilogarithms: Normalization Exceptions, Effective Degree Bounds, and an Infinitesimal Rigidity Criterion

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Abstract

We study the raw finite quantum-dilogarithm equations introduced by Radchenko–Wheeler. We classify their small-order normalization exceptions, compute the complete saturated solution schemes for the cyclic three-point metric and the anisotropic four-point metric, and give exact certificates for their reducedness. Using the known finiteness theorem, a Fourier–Weil eigenspace reduction gives explicit algebraic-degree bounds, with a sharper bound in prime order. Separately, we formulate a nonreduced-base realization hypothesis for the categorical construction and prove that it implies finite étaleness of the full raw solution scheme. The required flatness and coordinate-reconstruction arguments are distinguished from that hypothesis. An elementary calculation also recovers a known obstruction for higher-rank elementary abelian two-groups.

Status and provenance. This is a research draft. The exact small-order certificates and degree bounds do not depend on the general finite-étale criterion. That criterion is conditional on Hypothesis 3.1; it is not asserted here as an unconditional reducedness theorem. The original working note was dated September 21, 2026. This release compares it with Radchenko–Wheeler’s September 29 version [2], which restricts Theorem 5 to $N > 4$ and explicitly records small-order exceptions in Remark 3. The finite pentagon, algebraicity, and finiteness results are due to Radchenko–Wheeler, not to this note. The recent all-rank twisted-convolution theorem and modular-cocycle dictionary of Appleby–Flammia–Kopp [3] are likewise cited background. No priority claim is made for observing the small-order exceptions.

The companion functional paper [4] treats a different question: relations between quantum polylogarithms at fixed irrational parameter. Neither paper proves a numerical specialization theorem, Stark reciprocity, or the existence of the required unitary Galois conjugates.

1 The raw solution scheme

Let G be a finite abelian group of order $N > 2$, with an even quadratic function $q : G \rightarrow \mathbb{C}^\times$, and put

$$\chi(x, y) = \frac{q(x + y)}{q(x)q(y)}.$$

Assume χ is a nondegenerate bicharacter. Write $s = \sqrt{N} > 0$ and

$$\widehat{f}(u) = \frac{1}{s} \sum_{v \in G} f(v) \chi(-u, v).$$

All values of q are roots of unity. Choose a number field $K \subset \mathbb{C}$ containing these values and s .

The equations used throughout are

$$\chi(x, y) E(x) E(y) = C + \frac{1}{s} \sum_{t \in G} q(t) E(x - t) E(t) E(y - t), \quad C \neq 0. \quad (1)$$

This is the raw equation of [1, Definition 4, Eq. (29)], with its bicharacter retained. Our equation, rather than any version-dependent equation number, fixes the scheme studied below. For $N \leq 4$ we do not impose the extra zero-coordinate normalization used to extend the definition in [2, Remark 3]; thus our raw scheme includes the exceptional points. Set

$$P_{xy} = s(\chi(x, y) X_x X_y - C) - \sum_t q(t) X_{x-t} X_t X_{y-t},$$

$$X_{G,q} = \text{Spec } K[X_g : g \in G, C, C^{-1}] / (P_{xy} : x, y \in G).$$

Equivalently one can introduce T and add $TC - 1$.

The known finiteness of complex solutions [2, Section 5] does not imply reducedness of this defining ideal. We keep the actual localized ideal, rather than replacing it by its radical, when discussing tangent spaces and finite étaleness.

2 Small-order normalization exceptions

The normalization formulas in [2, Theorem 5], valid for $N > 4$, are

$$e^2 = se + 1, \quad e = E(0), \quad (2)$$

$$E(u)E(-u) = q(u)^{-1} \quad (u \neq 0), \quad (3)$$

$$\widehat{E}(u) = \lambda q(u) E(u), \quad C = -\lambda e, \quad \lambda^3 = \Gamma := \frac{1}{s} \sum_x q(x)^{-1}. \quad (4)$$

At the two smaller orders under consideration, the raw equation has the exceptions described below; compare [2, Remark 3].

2.1 An exact four-point counterexample

Let $G = (\mathbb{Z}/2\mathbb{Z})^2$ and

$$q(0) = 1, \quad q(x) = -1 \quad (x \neq 0), \quad E = q, \quad C = -1.$$

In the order 0, (1, 0), (0, 1), (1, 1) the bicharacter matrix is

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Thus the metric is nondegenerate. Its quadratic function has autocorrelation

$$\frac{1}{2} \sum_t q(x+t)q(y+t) = 2\delta_{x,y}.$$

Since $q(t)^2 = 1$, the cubic sum in (1) is exactly this quantity, while the left side is $q(x+y) = 2\delta_{x,y} - 1$. Consequently all sixteen ordered equations hold, with nonzero $C = -1$. But $e = 1$ gives $e^2 - 2e - 1 = -2$, and for nonzero u the asserted reflection identity reads $1 = -1$.

The first version's proof used nonconstancy of $q(x)E(x)$ on $G \setminus \{0\}$ for $N > 2$; here it is identically one. The current version avoids this step at these small orders by its $N > 4$ hypothesis and separate normalization convention [2, Theorem 5 and Remark 3].

This counterexample is algebraic. It is not a counterexample to the paper's algebraicity claim for Stark special values. Those values satisfy the normalizations separately through Theorem 2 and Proposition 10 of [1].

2.2 The order-three exception

Let $G = \mathbb{Z}/3\mathbb{Z}$, $r = (-1 + i\sqrt{3})/2$, and $q(1) = q(2) = r$. Then

$$E(0) = \frac{\sqrt{3} - i}{2}, \quad E(1) = E(2) = -\frac{\sqrt{3} + i}{2}, \quad C = r^2 \quad (5)$$

satisfies all nine equations (1). Again $e^2 - \sqrt{3}e - 1 = -2$. Complex conjugation gives the other metric. These statements follow by direct substitution; the script checks them in an exact number field, with no numerical tolerances.

2.3 Why there are no other normalization exceptions

Lemma 2.1. *For a complex solution of (1), all $E(u)$ and $\widehat{E}(u)$ are nonzero. If $q(u)E(u)$ is nonconstant on $G \setminus \{0\}$, then (2)–(4) hold. This implication also holds over the dual numbers in a neighbourhood of such a solution.*

Proof. We indicate explicitly the local-algebra issue. The Fourier transform of (1), with $p_x(z) = E(x-z)E(z)q(z)$, is

$$E(x)\widehat{E}(w-x) = C s \delta_0(w) + \widehat{E}(w)\widehat{p}_x(w). \quad (6)$$

The support argument and Fourier inversion in [1, proof of Theorem 5, (31)–(33)] give nonvanishing, and give

$$E(u)E(-u)q(u) = s e \delta_0(u) - C/\widehat{E}(0), \quad \widehat{E}(u) = \lambda q(u)E(u) \quad (u \neq 0).$$

The next convolution identity in that proof gives constants c, k such that $qE - c\widehat{E} = k\mathbf{1}$. For distinct nonzero u, v with $q(u)E(u) \neq q(v)E(v)$, subtraction yields $c\lambda = 1$, then $k = 0$. This extends the Fourier relation to zero. The algebra in the cited proof then yields

$$e^2 = (e^2 - se)(se + 1),$$

whose difference from the left side is $se(e^2 - se - 1)$. Since $se \neq 0$, this gives (2); reflection and $C = -\widehat{E}(0)$ follow, and the Fourier–Weil cubic gives $\lambda^3 = \Gamma$.

For the local assertion, let $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$ and reduce to the given solution. Every $E(u)$, $\widehat{E}(u)$ and C is now a unit. The chosen nonzero difference $q(u)E(u) - q(v)E(v)$ is also a unit. The Fourier and convolution calculations just described are algebraic identities and use only these invertible quantities and nonzero scalar constants. Thus the same argument works in A ; it does not infer reducedness from an identity that holds only at field-valued points. $\square$

Suppose now that $q(u)E(u) = B$ for all $u \neq 0$. Then $B \neq 0$ and

$$E = Bq^{-1} + (e - B)\delta_0.$$

The Gaussian Fourier identity $\widehat{q^{-1}} = \Gamma q$ shows that, for $u \neq 0$,

$$\widehat{E}(u) = B\Gamma q(u) + (e - B)/s.$$

But the partial Fourier relation obtained before the problematic sentence in [1] says $\widehat{E}(u) = \lambda B$ on this punctured group. Since $B\Gamma \neq 0$, the value $q(u)$ must be a constant r for $u \neq 0$. Orthogonality of any nontrivial row of χ gives

$$0 = 1 + r^{-2} + (N - 2)r^{-1}, \quad r + r^{-1} = -(N - 2).$$

As $|r| = 1$, this forces $N \leq 4$. For $N = 3$ it gives $r^2 + r + 1 = 0$. For $N = 4$ it gives $r = -1$, and the group must be $(\mathbb{Z}/2\mathbb{Z})^2$: a cyclic group of order four would have $q(2g) = q(g)^4 = 1$, a contradiction. The exact calculations in Section 5 show that the only constant-puncture solutions in these metrics are the two types already listed. We have proved the following corrected alternative.

Proposition 2.2. *Every complex solution of (1), with $N > 2$, either satisfies (2)–(4), or is the order-three solution (5) (or its conjugate), or the displayed four-point solution. In particular the normalization in Theorem 5 of [1] is valid without exceptions when $N \geq 5$, as in [2, Theorem 5].*

3 An infinitesimal criterion for finite étaleness

Hypothesis 3.1 (Normalized realization over dual numbers). *Let $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$ and let (E_A, C_A) be an A -valued solution of the raw equations reducing to a nonexceptional complex solution. After the normalization of Lemma 2.1, the Leavitt formulas, split fusion maps, and duality maps of [2, Section 5.4 and Appendix B] define the same identities over A . The scalar reconstruction of [2, Section 5.5] is compatible with this base change and with labelled monoidal equivalences reducing to the identity.*

The formulas have algebraic coefficients and unit denominators, which makes this a concrete ring-level verification problem. Field-valued identities alone do not establish the hypothesis. In particular, the extension to fields mentioned in [2, Remark 5] does not include nonreduced rings. We isolate this input rather than infer it from a zero-dimensional complex solution set. The finite computations below do not verify it for arbitrary metric groups.

Theorem 3.2 (Conditional finite-étale criterion). *Assume Hypothesis 3.1 for every nonexceptional complex solution. Then $X_{G,q}$ is finite étale over K for every metric group of order $N > 2$.*

The conclusion concerns the actual defining ideal, including infinitesimal structure. The proof below supplies the flatness and rigidity arguments once the specified realization input is available.

3.1 The linearized equations

At a nonexceptional solution write

$$E_A(u) = E(u) + \varepsilon h(u), \quad C_A = C + \varepsilon c_1, \quad A = \mathbb{C}[\varepsilon]/(\varepsilon^2).$$

Lemma 2.1 applies over A . Consequently

$$(2e - s)h(0) = 0, \quad 3\lambda^2\lambda_1 = 0.$$

The coefficients are nonzero: $(2e - s)^2 = N + 4$ and $\lambda \neq 0$. Hence $h(0) = \lambda_1 = c_1 = 0$. The remaining tangent equations include

$$E(-u)h(u) + E(u)h(-u) = 0 \quad (u \neq 0), \quad (7)$$

$$\hat{h}(u) = \lambda q(u)h(u), \quad (8)$$

and

$$\begin{aligned} \chi(x, y)(h(x)E(y) + E(x)h(y)) &= \frac{1}{s} \sum_t q(t)(h(x-t)E(t)E(y-t) \\ &\quad + E(x-t)h(t)E(y-t) \\ &\quad + E(x-t)E(t)h(y-t)). \end{aligned} \quad (9)$$

It is not sufficient merely to observe that there are many equations. Under Hypothesis 3.1, we show that this system has zero kernel by reconstruction from a flat categorical deformation.

3.2 Extending the construction over the dual numbers

Put $d = -s/e$ and fix a square root of d . These parameters are constant in A , since e is constant. Form the Leavitt algebra $L_A = A \otimes_{\mathbb{C}} L_{2N}$. By Hypothesis 3.1, the formulas of [2, Section 5.4] define A -linear endomorphisms α_g and ρ_A of L_A , together with the stated split fusion and duality maps. Their coefficients are algebraic expressions in $E_A, q, s, \lambda, d^{1/2}$; the denominators are units. The identity verification in the hypothesis is used precisely here, before any appeal to deformation rigidity. In particular,

$$\rho_A^2 \simeq \bigoplus_{g \in G} \alpha_g \oplus \rho_A^{\oplus N}, \quad \alpha_g \rho_A = \rho_A, \quad \rho_A \alpha_g \simeq \rho_A.$$

Flatness of morphism spaces. This is the additional check that cannot be replaced by saying simply that “the formulas work over A .” For the basic objects α_g, ρ_A , the reduced category has scalar endomorphisms and zero morphisms between distinct simples, as computed in [2, Section 5.4]. Let $z = z_0 + \varepsilon z_1$ be an intertwiner in L_A . Between distinct basic simples reduction gives $z_0 = 0$, and the remaining intertwining equation says that z_1 is an intertwiner between the corresponding distinct reduced simples. Thus $z_1 = 0$. For an endomorphism of one basic simple, z_0 is a scalar. Subtract that scalar identity, which is an exact intertwiner over A ; the same reasoning shows that z_1 is a scalar. Therefore

$$\text{End}_A(\alpha_g) = A, \quad \text{End}_A(\rho_A) = A, \quad \text{Hom}_A(S, T) = 0 \quad (S \not\simeq T)$$

for these chosen basic simples. The explicit fusion decompositions extend this to all tensor words as free finite-rank A -modules. Choosing the split skeleton and bases of the fusion spaces therefore gives a flat deformation of the reduced fusion category, with the same fusion rules.

First-order rigidity. The theorem of Etingof–Nikshych–Ostrik gives $H^3(\mathcal{C}) = 0$ for a fusion category in characteristic zero. This cohomology classifies first-order deformations of the associativity structure [5, Theorem 2.27, §7.1]. It follows that the deformation just constructed is monoidally equivalent, with its reduction fixed, to the constant deformation $A \otimes_{\mathbb{C}} \mathcal{C}$.

3.3 Why gauge-triviality forces the actual coordinates to be fixed

The preceding paragraph alone does not show $h = 0$: gauge-trivial structure constants can still move. We use the scalar reconstruction of [2, Section 5.5] over A , as required in Hypothesis 3.1. There the invariant line determining $T_0 \in \text{Hom}(\rho, \rho^2)$ is the trivial-character line for the α_g action. Over A it is still free of rank one, since the distinct character values have invertible differences. The normalization of U_g by its action on the rank-one space $\text{Hom}(1, \rho^2)$ works over A as well. Setting $T_g = U_{-g}T_0$ gives a basis up to multiplication by a common unit of A . Let T'_g be the dual basis. The reconstruction identity is

$$\frac{E_A(g)}{s} 1_\rho = T'_0(T'_0 \otimes 1_\rho)(1_\rho \otimes T_g)T_0. \quad (10)$$

All tensor associativity identifications are understood in this expression. The two basis factors and two dual-basis factors cancel the common rescaling. Thus the right side is a scalar invariant of the labelled monoidal category, not a coordinate depending on a moving basis. Under the equivalence to the constant deformation it is therefore $E(g)/s$. Relabelling by $\text{Aut}(G)$ cannot create a first-order motion of labels; the equivalence is chosen to have fixed reduction. Equation (10) gives $E_A(g) = E(g)$, hence $h(g) = 0$ for all g .

3.4 Finishing the scheme argument

Under Hypothesis 3.1, every nonexceptional complex point has $T_E X_{G,q} = 0$. At the order-three exception, the Jacobian of the residuals P_{xy} in variables $(E(0), E(1), E(2), C)$ has a nonzero 4×4 minor

$$-\frac{27\sqrt{3}}{2} - \frac{81i}{2}.$$

At the four-point exception it has a 5×5 minor equal to 512. These calculations are certified by the accompanying script; alternatively the saturated ideals in Section 5 establish reducedness there. Thus every complex point of $X_{G,q}$ has zero tangent space.

For an affine scheme of finite type over $\mathbb{C}$, zero tangent space at every closed point implies dimension zero. In each local ring its maximal ideal satisfies $\mathfrak{m}/\mathfrak{m}^2 = 0$, hence is zero by Nakayama's lemma. The scheme is therefore a finite reduced complex scheme. It follows by base change and descent that $X_{G,q}$ is finite etale over K . This proves Theorem 3.2. Notice that this argument uses the infinitesimal part of Ocneanu rigidity; separate use of its finiteness-of- categories conclusion is unnecessary.

Corollary 3.3. *Under the hypothesis of Theorem 3.2, the localized ideal defining $X_{G,q}$ is radical. For every local Artin $\mathbb{C}$ -algebra A with residue field $\mathbb{C}$, reduction induces a bijection $X_{G,q}(A) \rightarrow X_{G,q}(\mathbb{C})$. In particular, there are no hidden nilpotent motions of a fixed solution of any order.*

4 Effective degree bounds from Fourier–Weil eigenspaces

The results of this section use the established finiteness theorem [2, Section 5], not Hypothesis 3.1 or reducedness. Fix choices e, λ satisfying (2) and $\lambda^3 = \Gamma$, and write $K' = K(e, \lambda)$. Define the Fourier–Weil operator and its eigenspace dimension by

$$Wf(u) = q(u)^{-1} \hat{f}(u), \quad m_\lambda = \dim_{K'} \ker(W - \lambda).$$

The identity $W^3 = \Gamma I$ is proved in [1, proof of Theorem 5]. In this fixed sector every normalized solution lies in

$$\ker(W - \lambda) \cap \{X_0 = e\}.$$

If the sector is nonempty, this is an affine space of dimension $m_\lambda - 1$; the functional $X \mapsto X_0$ is then nonzero on the eigenspace.

Proposition 4.1. *For a normalized finite quantum dilogarithm in a nonempty (e, λ) sector,*

$$[K'(E(g) : g \in G) : K'] \leq 3^{m_\lambda - 1}.$$

When $G = \mathbb{Z}/p\mathbb{Z}$ with p prime, the sharper bound is

$$[K'(E(g) : g \in G) : K'] \leq 2^{m_\lambda - 1}.$$

The same bounds hold for the number of geometric solutions in that sector.

Proof. In the affine eigenspace section the remaining defining equations have degree at most three. Its finite solution set has at most $3^{m_\lambda - 1}$ points by the affine Bezout bound. The field degree is the size of the orbit of the entire tuple under the absolute Galois group of K' , so it is bounded by that number. In prime order, [2, Section 5.1, prime-order uncertainty argument] shows that the larger variety cut out just by the linear Fourier equations and quadratic reflection identities is finite: at infinity, a vector and its Fourier transform would have equal skew support of size at most $(p - 1)/2$, contrary to the prime-order uncertainty principle. Consequently quadratic, rather than cubic, Bezout applies on the same affine space. $\square$

The dimensions m_λ are explicitly computable from a finite matrix. They also admit a uniform estimate.

Lemma 4.2. *Let $G[3] = \{g : 3g = 0\}$. Then*

$$m_\lambda \leq \left\lfloor \frac{N + 2\sqrt{|G[3]|}}{3} \right\rfloor.$$

Proof. The evenness of q implies $\chi(x, x) = q(x)^2$ and $q(nx) = q(x)^{n^2}$. Therefore

$$\mathrm{tr} W = \frac{1}{s} \sum_x q(x)^{-3}.$$

The radical of χ^3 is $G[3]$. For $z \in G[3]$, $q(z)^9 = q(z)^6 = 1$, so $q(z)^3 = 1$. The usual direct calculation of the squared absolute value of a Gaussian sum thus gives

$$\left| \sum_x q(x)^{-3} \right|^2 = N|G[3]|, \quad |\mathrm{tr} W| = \sqrt{|G[3]|}.$$

For completeness, expand the square using $x = y + z$, sum first over y , and use character orthogonality; only $z \in G[3]$ survive.

Choose $\lambda_0^3 = \Gamma$ and $\omega = e^{2\pi i/3}$. Since $W^3 = \Gamma I$ and $|\Gamma| = 1$, the eigenvalues are $\lambda_0 \omega^j$ with multiplicities m_j . If $T = \mathrm{tr}(W)/\lambda_0$, Fourier inversion on three points gives

$$m_j = \frac{N + 2\mathrm{Re}(\omega^{-j}T)}{3}.$$

The asserted bound follows. $\square$

Corollary 4.3. *For $p \geq 5$ prime, a fixed metric q on $\mathbb{Z}/p\mathbb{Z}$, and $K \supset \mathbb{Q}(q(g), \sqrt{p})$,*

$$\#X_{G,q}(\overline{K}) \leq 6 \cdot 2^{\lfloor (p-1)/3 \rfloor}, \quad [K(E(g) : g \in G) : K] \leq 6 \cdot 2^{\lfloor (p-1)/3 \rfloor}.$$

Proof. There are at most two choices for e and three for λ . Here $G[3] = \{0\}$, so $m_\lambda - 1 \leq \lfloor (p-1)/3 \rfloor$. There are no Gaussian exceptions for $p \geq 5$. $\square$

These are deliberately coarse explicit upper bounds, not predictions of the actual class-field degrees. Sector-specific dimensions are usually more useful.

5 Complete small-order certificates

5.1 The cyclic group of order three

Use $G = \mathbb{Z}/3\mathbb{Z}$, $s = \sqrt{3}$, $r = (-1 + is)/2$, $q(1) = q(2) = r$, and $K = \mathbb{Q}(s, i) = \mathbb{Q}(\zeta_{12})$. Write $E = (a, b, d)$ and $C = c$. Eliminating the ideal $(P_{xy}, tc - 1)$ with lexicographic order t, c, d, a, b gives

$$K[X_{G,q}] \simeq K[b]/(p(b)),$$

where

$$\begin{aligned} p(b) &= b^5 + 2ib^4 - 2b^3 + (s - i)b^2 + (1 + si)b - (s - i)/2 \\ &= \left(b + \frac{s + i}{2}\right) Q(b), \\ Q(b) &= b^4 + \frac{-s + 3i}{2}b^3 - \frac{1 + si}{2}b^2 + sb + r. \end{aligned}$$

The other coordinates are recovered by

$$\begin{aligned} a &= -ib^4 + \frac{5 + si}{2}b^3 - (s - 2i)b^2 - sib + \frac{3s + i}{2}, \\ d &= ib^4 - (1 + si)b^3 + (s + i)b^2 - (1 - si)b - 2i, \\ c &= -(1 - si)b^4 - \frac{3s + 5i}{2}b^3 + (2 - si)b^2 - (s - 3i)b - \frac{5 + si}{2}, \\ t &= sib^4 - (2s + 2i)b^3 + (3 - si)b^2 + 3ib - \frac{3 + si}{2}. \end{aligned}$$

These polynomials, together with $p(b)$, form the computed reduced Groebner basis in solved form. The script regenerates it directly from the original cubic equations, rather than accepting these formulas as input.

Exact discriminant calculations give

$$\text{disc } Q = -147, \quad Q\left(-\frac{s + i}{2}\right) = r, \quad \text{disc } p = -147r^2 \neq 0.$$

Thus there are exactly five reduced geometric points. The linear factor gives (5). The four roots of Q give the normalized solutions, with $\lambda = (s - i)/2$; substitution modulo Q verifies all of (2)–(4).

One can also see that no normalized solution has $b = d$. In the normalized part the elimination equations give

$$b + d = \frac{a(1 - si)}{2}, \quad (1 + si)b^2 - 2ab + 1 - si = 0, \quad a^2 - sa - 1 = 0.$$

The discriminant of the quadratic in b is $4(a^2 - 4)$, which is nonzero under the equation for a . Hence only the exceptional point has a constant punctured value, proving the needed order-three part of Proposition 2.2.

The coefficients of p are algebraic integers in K . Consequently $\mathcal{O}_K[1/21][b]/(p)$ is an explicit finite etale model of this generic solution algebra. This is a concrete good-reduction statement outside primes above 3 and 7 for this model, not a general conductor theorem for Stark values. The primitive polynomial gives an algebraic model; identifying its points with analytically selected special values is a separate task.

5.2 The anisotropic four-point metric

For $G = (\mathbb{Z}/2\mathbb{Z})^2$, $q = (1, -1, -1, -1)$, $E = (a, b, d, e)$, the reduced lexicographic Groebner basis of $(P_{xy}, tc - 1)$ in the order t, c, e, d, b, a is

$$t + 1, \quad c + 1, \quad e + 1, \quad d + 1, \quad b + 1, \quad a - 1.$$

Thus this entire metric-specific solution scheme is the single reduced rational point already displayed. The existence theorem in [1] is an existence statement for each underlying two-generator group, not for every quadratic function on that group. There is no contradiction with that theorem.

6 An elementary recovery of a known higher-rank obstruction

Schopieray's earlier classification [6, Theorem 1.1] implies that the Izumi near-group ring for $(\mathbb{Z}/2\mathbb{Z})^r$ with multiplicity 2^r is not categorifiable for $r \geq 3$. Thus the nonexistence conclusion below is *not new as a categorical consequence*. It is useful to recover it straight from the finite equations, without categorical rigidity or a unitarity assumption.

Proposition 6.1. *For every $r \geq 3$ and every metric quadratic function on $G = (\mathbb{Z}/2\mathbb{Z})^r$, equation (1) has no solution with $C \neq 0$.*

Proof. Put $N = 2^r$. Since $N \geq 8$, the corrected normalization has no exception. Every $u \neq 0$ is its own negative, so

$$q(u)^4 = 1, \quad E(u)^2 = q(u)^{-1}, \quad E(u) \in \mu_8.$$

Let $L = \mathbb{Q}(\zeta_8)$. The value $s = \sqrt{N}$ lies in L , and the Fourier matrix has entries $\pm 1/s$. Since qE is nonconstant off zero, subtracting the Fourier equations at two nonzero points with distinct values of qE shows that $\lambda \in L$: the terms involving $E(0)$ cancel and all other terms lie in L . Either Fourier equation then gives $e = E(0) \in L$. Because the two possible values of e are real, $e \in \mathbb{Q}(\sqrt{2})$.

The identity $2e - s = \pm\sqrt{N+4}$ implies $\sqrt{2^{r-2}+1} \in \mathbb{Q}(\sqrt{2})$. The positive integer $2^{r-2}+1$ is odd. An odd integer whose square root belongs to $\mathbb{Q}(\sqrt{2})$ must be a square in $\mathbb{Z}$: its square class cannot be 2. Thus

$$2^{r-2} + 1 = m^2.$$

The consecutive even integers $m-1, m+1$ have product a power of two. They must be 2, 4, so $m = 3$, $r = 5$, and $N = 32$.

It remains to exclude that order. Write $z = \zeta_8$ and $\sqrt{2} = z - z^3$. The Gaussian sum $\sum q(u)^{-1}$ is a Gaussian integer of norm 32. It is therefore one of $\pm 4 \pm 4i$, so Γ is an odd power of z . As $\lambda^3 = \Gamma$ and $\lambda \in L$, it follows that λ is an odd power of z as well. Here the only roots of unity in L are μ_8 ; equivalently, a nontrivial cube root of unity would generate the quadratic subfield $\mathbb{Q}(\sqrt{-3})$, which L does not have.

The Fourier equation at zero now requires a sum of 31 eighth roots of unity to equal

$$\sum_{u \neq 0} E(u) = (4\sqrt{2}\lambda - 1)(2\sqrt{2} \pm 3).$$

Use the rational basis $1, z, z^2, z^3$ of L . Any sum of 31 elements of $\mu_8 = \{\pm 1, \pm z, \pm z^2, \pm z^3\}$ has coefficient ℓ^1 norm at most 31. For $\lambda = z, z^3, z^5, z^7$ the displayed right side has coefficient norm, respectively,

$$37, \quad 47, \quad 47, \quad 37,$$

for either sign. For example when $\lambda = z$ and the sign is positive it is $9 + 14z + 12z^2 + 2z^3$. The other seven expansions are included in the exact verification output. This contradiction excludes $N = 32$ as well. $\square$

7 Arithmetic scope

Under Hypothesis 3.1, Theorem 3.2 gives a finite étale K -algebra, hence a product of finite separable extensions. For the two explicitly computed metrics in Section 5, this conclusion is unconditional. A generic linear combination of the coordinate functions is a primitive element, yielding a squarefree univariate presentation and polynomial recovery of every coordinate. Clearing denominators in this presentation and its inverse, and inverting its discriminant, gives an effectively computable integer $D \neq 0$ and a finite étale model over $\mathcal{O}_K[1/D]$. This provides good reduction and unique infinitesimal lifting outside a finite computable set; no small uniform formula for D is claimed.

These conclusions do not prove algebraic integrality or a unit property, identify the predicted ray class fields, determine the Artin action, or produce unitary Galois conjugates. Algebraicity and the equations already gave a Galois permutation action. Finite étaleness adds infinitesimal and reduction control, not an automatic reciprocity law. These distinctions agree with the open questions in [1, §6].

For an EIT-style program the contribution is methodological: study the actual relation scheme and its tangent spaces, rather than only numerical relations at one point. No polynomial EIT theorem is applied here to the quantum dilogarithm; its categorical relation system is analysed on its own terms.

Reproducibility and research disclosure

The script `scripts/verify_rw_extensions.py` regenerates the exceptional solutions, full-rank tangent minors, and saturated low-order Gröbner bases from the raw equations. It also checks the finite root-of-unity calculation for $N = 32$. Its output is recorded in the repository. These exact checks prove the stated finite certificates, not the arbitrary-group realization hypothesis. AI-assisted tools were used for drafting, proof exploration, literature comparison, and internal checks. The manuscript has not been independently refereed or formally verified.

References

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